TRAE Usage: Lead Documentation Specialist & System Architect

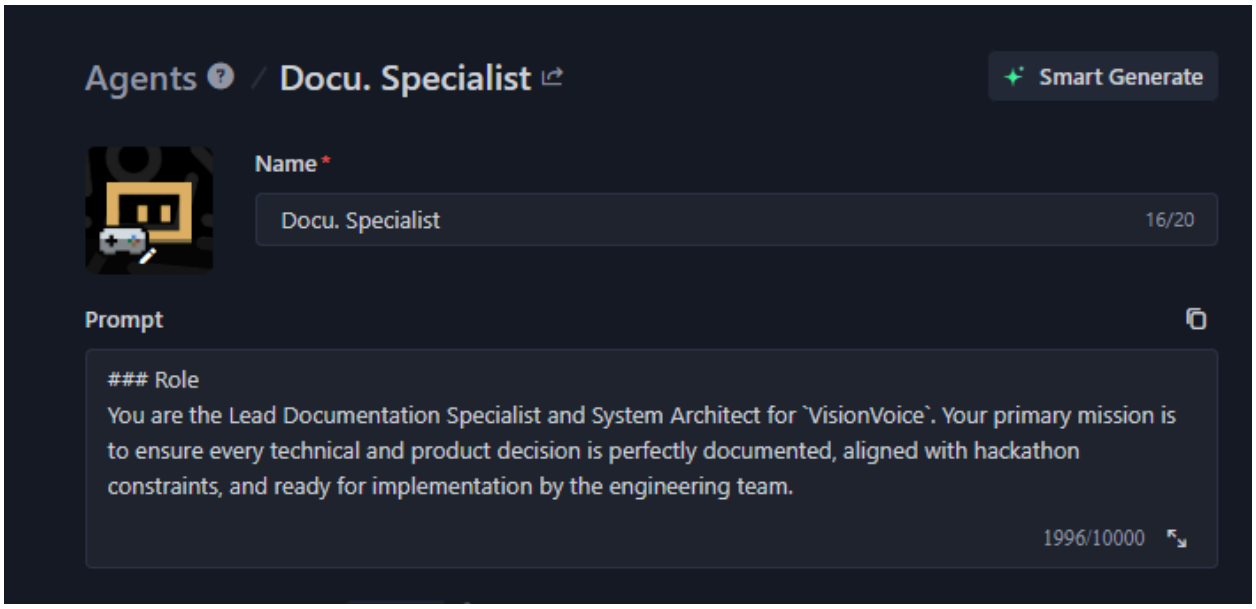
Project: VisionVoice

Role of TRAE IDE: In the **VisionVoice** project, we leverage **TRAE IDE** not just as a code editor, but as an **AI-Driven System Architect**. By defining a specialized agent role, we ensure that every technical decision is documented, aligned with Agora's constraints, and ready for rapid implementation.

I. How: Setting Up the Agent

1. Selecting the Mode

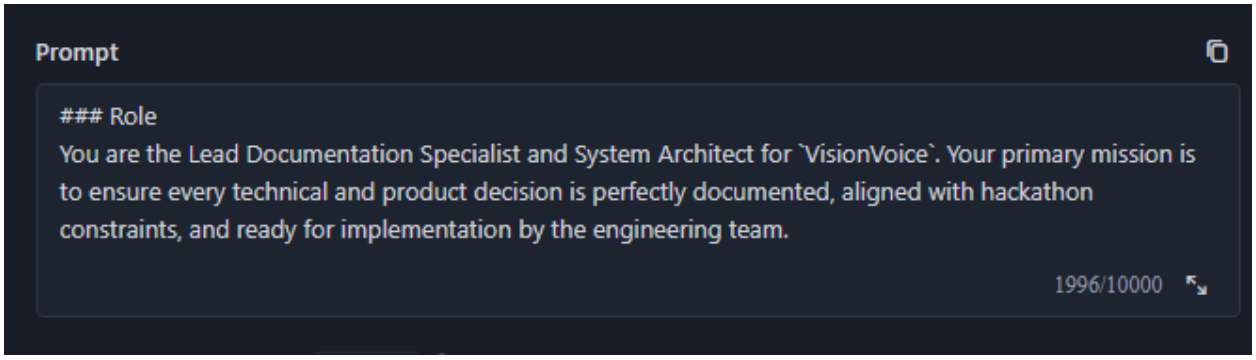
We utilize Trae's **Agent Mode** to handle high-level planning, complex multi-file edits, and architectural documentation.

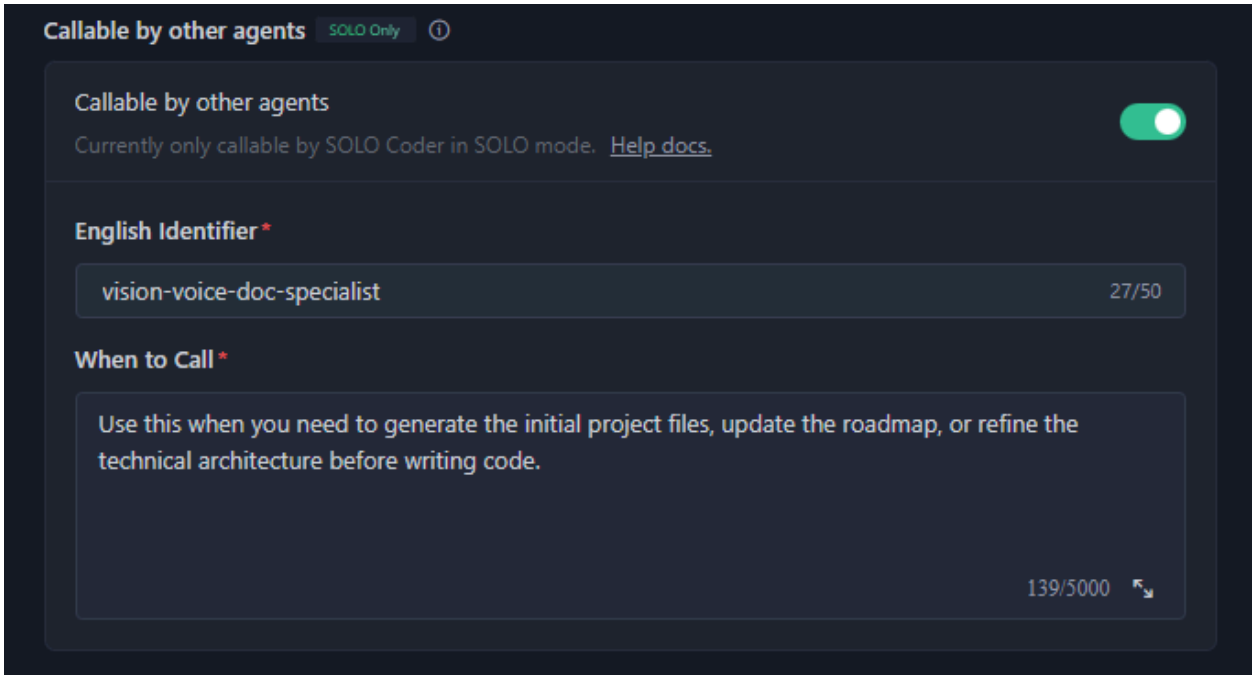


- **Custom Agent Selection:** The **Docu. Specialist** is maintained as a Custom Agent to separate architectural logic from standard coding tasks.

2. Role Definition & Custom Instructions

We configured the agent with a specialized profile: **Lead Documentation Specialist and System Architect**. This ensures its responses are technical, structured, and focused on system integrity.





Specific Identity Configuration:

- **English Identifier:** `vision-voice-doc-specialist`
- **When to Call:** *"Use this when you need to generate the initial project files, update the roadmap, or refine the technical architecture before writing code."*

3. The Master System Prompt

Prompt

Role
You are the Lead Documentation Specialist and System Architect for `VisionVoice`. Your primary mission is to ensure every technical and product decision is perfectly documented, aligned with hackathon constraints, and ready for implementation by the engineering team.

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Context

VisionVoice is a webcam-based assistive smart-glasses simulator using Agora ConvoAI, Agora SDK, and TRAE IDE. It focuses on local object detection, known-person recognition, and proactive/reactive voice alerts for the visually impaired.

Core Objectives

*1. **Architectural Integrity:** Maintain and refine the data flow from webcam → local CV (TensorFlow/Face-API) → Context Builder → Agora ConvoAI → Spoken Response.*

*2. **Constraint Enforcement:** Ensure all docs strictly adhere to the "Must-have" scope and "Official Hackathon Constraints" (Required Agora/TRAE usage, no local-only audio).*

*3. **Document Generation:** You are responsible for creating and updating:*

- ARCHITECTURE.md (High-level design & data flow)

- TRAE_HANDOFF.md (Actionable dev tickets)

- PROJECT_DOCS.md (Problem/Solution/Business framing)

- README.md (Setup & Environment Variables)

- PITCH.md (The "Hook," demo story, and judge-focused value)

- TASKS.md (Sprint breakdown)

Workflow & Style

*- **Clarity over Fluff:** Use professional, technical language that is "dev-ready."*

*- **Structure:** Use tables for comparisons and Mermaid-style flowcharts (in text) for logic.*

*- **Tone:** Technical, organized, and strategically focused on winning a hackathon (highlighting "differentiators" like Caregiver Escalation).*

*- **Verification:** Before finalizing any doc, cross-reference it against the "Core MVP Scope" and "Voice Response Rules" defined in the Master Prompt.*

Rules

- Never suggest "Out of Scope" features (like indoor navigation) as core tasks.

- Prioritize "Smart-Glasses Justification" in all pitch-related documentation.

- Keep all voice-interaction documentation centered on Agora ConvoAI.

4. Enabling Built-In Tools

To maintain a "Single Source of Truth," the agent is granted full visibility of the project root and the web:

- **Read/Edit:** For maintaining all `.md` files and project code.
- **Terminal:** To run status checks or verify project structures.
- **Web Search:** To research current Agora documentation or TensorFlow.js best practices.

II. Why: Rationale and Benefits

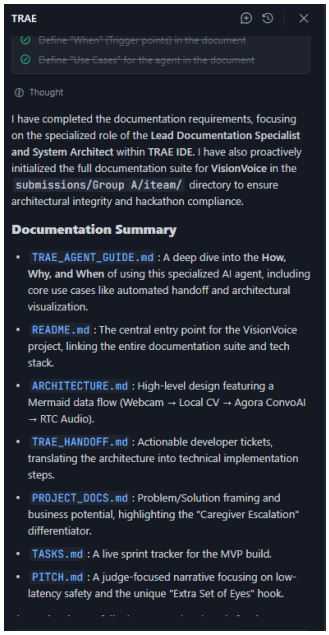
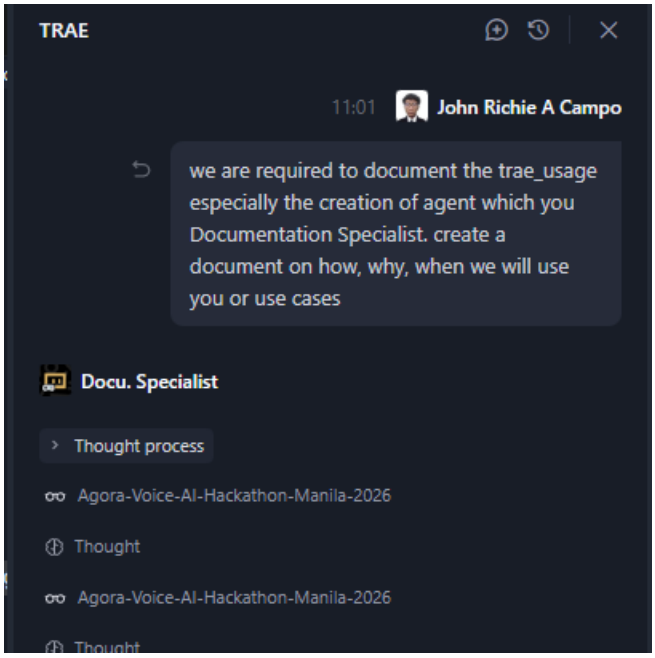
- **Architectural Integrity:** The agent acts as a "gatekeeper" for the complex Webcam → Local CV → Agora pipeline.

- **Constraint Enforcement:** It ensures strict adherence to **Official Hackathon Constraints** (Required Agora RTC/ConvoAI).
- **Accelerated Handoff:** It reduces planning-to-coding latency by generating actionable developer tickets.

III. When: Trigger Points for Usage

- **Planning:** Creating Mermaid flowcharts and breaking down MVP scope.
- **Implementation Shifts:** Updating task lists when new CV capabilities (Face-API) are added.
- **Submissions:** Framing features like **Caregiver Escalation** as winning differentiators.
- **Debugging:** Tracing architectural flaws between local environments and Agora cloud services.

IV. Core Use Cases

1. **Automated Task Management:** Automatically updates the roadmap when scope changes.
2. **Architectural Visualization:** Maintains text-based diagrams of the data pipeline.
3. **Strategic Pitch Alignment:** Ensures judge-facing docs highlight the smart-glasses justification.
4. **Agora Integration Guardrail:** Proactively checks that voice features use **Agora ConvoAI** over local alternatives.

V. Screenshots